Module 3 — Forces and Motion

This module covers kinematics, dynamics, momentum, energy, and power — the core mechanics of A-Level Physics.

3.1 Motion Graphs

Displacement-Time Graphs

- **Gradient** = velocity
- A horizontal line means the object is stationary
- A straight line with a positive gradient means constant velocity in the positive direction
- A curve means the velocity is changing (accelerating or decelerating)
- The gradient of the tangent to a curve at any point gives the instantaneous velocity
- A negative gradient means the object is moving in the negative direction

Velocity-Time Graphs

- **Gradient** = acceleration
- **Area under the graph** = displacement
- A horizontal line means constant velocity (zero acceleration)
- A straight line with a positive gradient means constant acceleration
- A straight line with a negative gradient means constant deceleration
- The area under the curve can be found by counting squares, using the trapezium rule, or geometric shapes
- Area below the time axis represents negative displacement (motion in the opposite direction)

Exam Tip: Be careful to distinguish between distance-time and displacement-time graphs. Distance can only increase or stay the same; displacement can decrease (return to start). Similarly, speed-time

graphs have no negative values; velocity-time graphs can.

3.2 SUVAT Equations

SUVAT equations apply only to motion with **constant (uniform) acceleration** in a straight line.

Variables: **s** = displacement, **u** = initial velocity, **v** = final velocity, **a** = acceleration, **t** = time.

$$v = u + at$$

$$s = ut + \tfrac{1}{2}at^2$$

$$s = vt - \tfrac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

$$s = \tfrac{1}{2}(u + v)t$$

Equation	Missing Variable
$v = u + at$	s
$s = ut + \tfrac{1}{2}at^2$	v
$s = vt - \tfrac{1}{2}at^2$	u
$v^2 = u^2 + 2as$	t
$s = \tfrac{1}{2}(u + v)t$	a

Exam Tip: To choose which equation to use, list the three known quantities and the one unknown. Pick the equation that contains exactly those four variables (the one that is missing the variable you do not know and do not need).

3.3 Projectile Motion

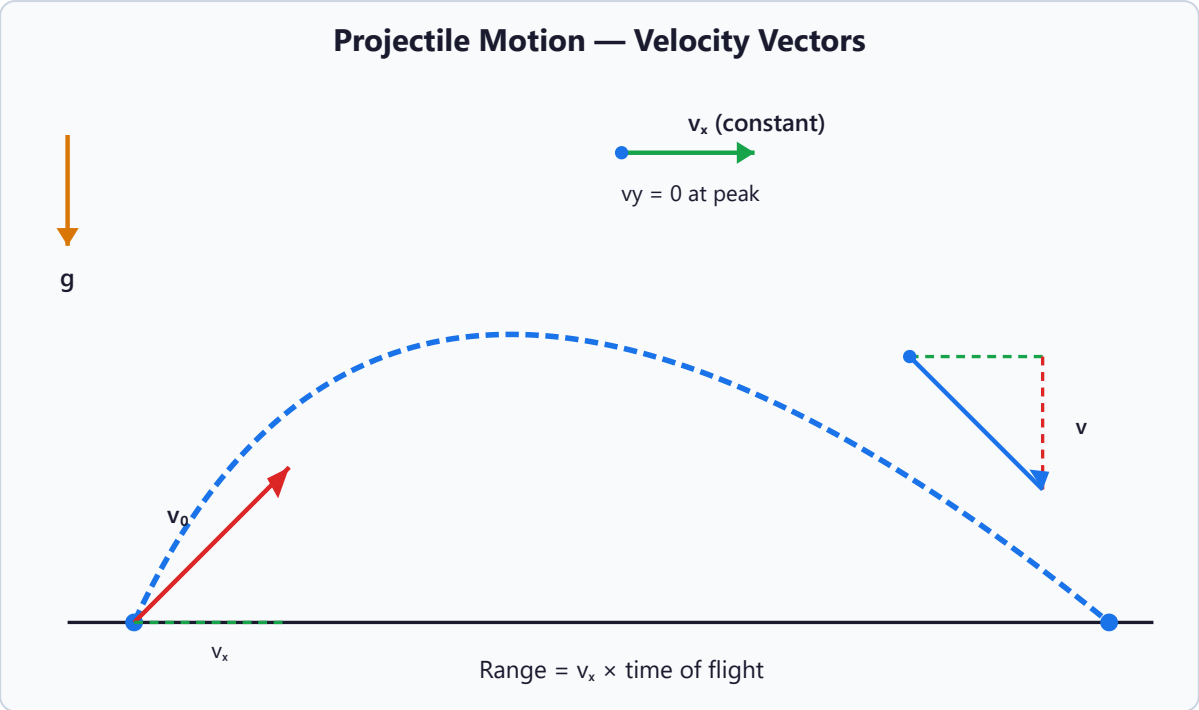
Projectile motion: The motion of an object under the influence of gravity alone (ignoring air resistance). The horizontal and vertical components of motion are independent.

- **Horizontal:** No acceleration (constant velocity). $x = v_x t$, where $v_x = v \cos \theta$
- **Vertical:** Constant acceleration $g = 9.81\ \text{m s}^{-2}$ downwards. Use SUVAT with $a = \pm g$
- At the highest point of a trajectory, the vertical velocity = 0 (but horizontal velocity remains unchanged)
- The time of flight is determined by the vertical motion
- The range is determined by the horizontal velocity multiplied by the time of flight

For a projectile launched at speed v at angle θ above the horizontal:

- Horizontal component of velocity: $v_x = v \cos \theta$ (constant throughout)
- Initial vertical component: $v_y = v \sin \theta$
- At any time t: vertical velocity $= v \sin \theta - gt$
- At any time t: horizontal displacement $= (v \cos \theta)t$
- At any time t: vertical displacement $= (v \sin \theta)t - \frac{1}{2}gt^2$

© RAG Learning · Not for redistribution



Exam Tip: Always resolve the initial velocity into horizontal and vertical components first. Then treat horizontal and vertical motions separately, linked only by the common variable time.

3.4 Newton's Laws of Motion

Newton's First Law: An object will remain at rest or continue to move at constant velocity unless acted upon by a resultant force.

Newton's Second Law: The resultant force on an object is equal to the rate of change of its momentum. For constant mass: F = ma, where F is the resultant force (N), m is mass (kg), and a is acceleration (m s^{-2}).

Newton's Third Law: If body A exerts a force on body B, then body B exerts an equal and opposite force on body A. The two forces are of the same type, act on different objects, and exist simultaneously.

Exam Tip: Newton's Third Law pairs must be: (1) the same type of force, (2) equal in magnitude, (3) opposite in direction, (4) acting on different objects. "Weight and normal contact force" is NOT a Newton's Third Law pair — they act on the same object.

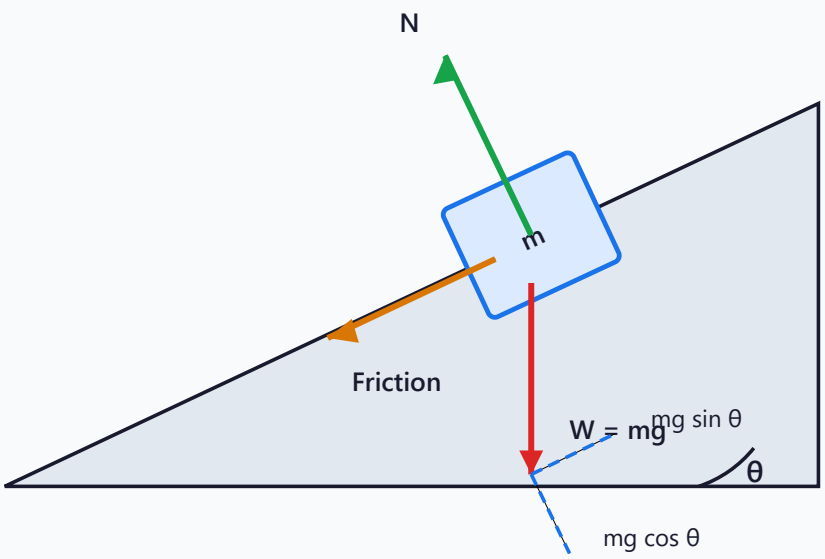
Weight

$$W = mg$$

Where W = weight (N), m = mass (kg), g = gravitational field strength (N kg^{-1} or m s^{-2}). On Earth's surface, g = 9.81 N kg^{-1}.

Free Body Diagrams

Free Body Diagram — Block on an Inclined Plane



A free body diagram shows all the forces acting on a single object as labelled arrows. The length of each arrow represents the relative magnitude. Forces to include:

- Weight (always vertically downward from the centre of mass)
- Normal contact / reaction force (perpendicular to the surface)
- Friction / drag (opposing the direction of motion)

- Tension (along a string or cable, away from the object)
- Applied / driving force

3.5 Friction and Drag

Friction: A contact force that opposes the relative motion (or attempted motion) between two surfaces.
Drag: A resistive force that opposes the motion of an object through a fluid (liquid or gas). Drag depends on speed, cross-sectional area, shape, and fluid density.

Terminal Velocity

When an object falls through a fluid:

1. Initially, the only force is weight, so the object accelerates at g
2. As speed increases, drag increases (drag is proportional to speed or speed^2)
3. The resultant force decreases, so acceleration decreases
4. Eventually, drag equals weight: resultant force = 0, acceleration = 0
5. The object now moves at a constant speed called **terminal velocity**

Exam Tip: A velocity-time graph for a falling object approaching terminal velocity shows a curve that starts steep and gradually flattens to a horizontal line. The gradient (acceleration) decreases over time.

3.6 Momentum

Momentum (p): The product of mass and velocity. It is a vector quantity.

$$p = mv$$

Units: kg m s^{-1} (or equivalently, N s)

Newton's Second Law (General Form)

$$F = \Delta p / \Delta t = (mv - mu) / \Delta t$$

The resultant force equals the rate of change of momentum. For constant mass, this reduces to F = ma.

Impulse

Impulse: The product of force and the time for which it acts. Impulse equals the change in momentum.

$$\text{Impulse} = F\Delta t = \Delta p = mv - mu$$

Units: N s (or kg m s^{-1})

- On a force-time graph, the impulse is the area under the graph
- A longer collision time means a smaller average force for the same impulse — this is the principle behind crumple zones, airbags, and crash mats

Conservation of Momentum

Law of Conservation of Momentum: In a closed system (no external forces), the total momentum before an interaction equals the total momentum after the interaction.

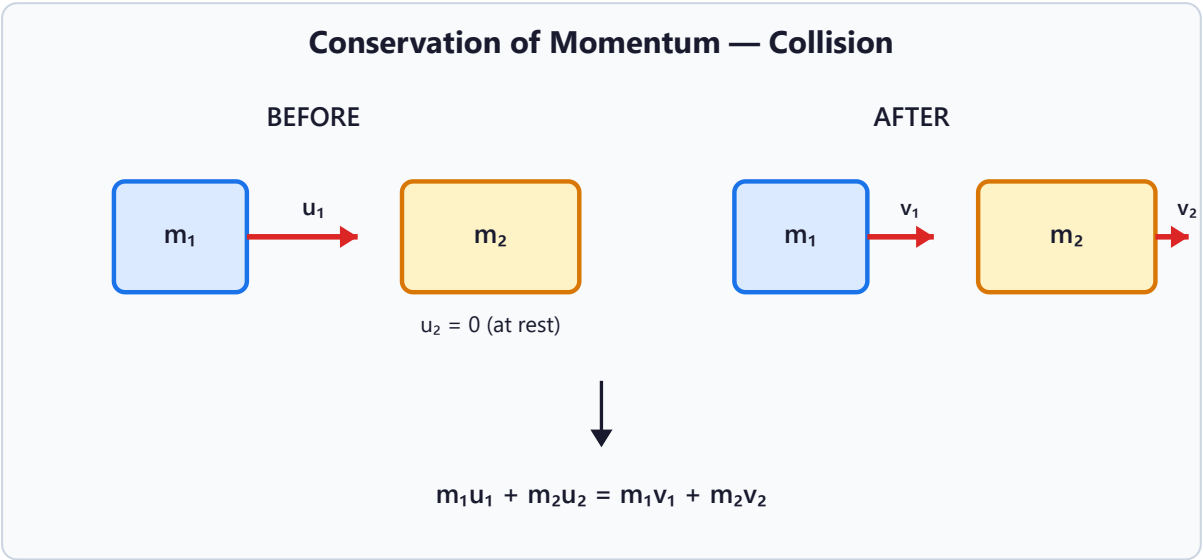
$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$

Types of Collision

Property	Elastic Collision	Inelastic Collision	Perfectly Inelastic
Momentum	Conserved	Conserved	Conserved
Kinetic Energy	Conserved	NOT conserved (some KE converted to other forms)	Maximum KE lost (objects stick together)

© RAG Learning · Not for redistribution

Total Energy	Conserved	Conserved	Conserved
Example	Colliding gas molecules	Most real collisions	Catching a ball; cars crumpling together



Exam Tip: Momentum is always conserved in collisions (provided the system is closed). To check if a collision is elastic, calculate total KE before and after — if they are equal, it is elastic. Explosions are the reverse of perfectly inelastic collisions.

3.7 Work, Energy, and Power

Work Done

Work done: The product of the force and the displacement in the direction of the force. Work done is equal to energy transferred.

$$W = Fd \cos \theta$$

Where W = work done (J), F = force (N), d = displacement (m), θ = angle between force and displacement.

- If θ = 0°: W = Fd (force and displacement in the same direction)

- If θ = 90°: W = 0 (force perpendicular to displacement — no work done)
- If θ = 180°: W = −Fd (force opposes displacement)
- 1 joule = 1 newton-metre: the work done when a force of 1 N moves through 1 m

Kinetic Energy

$$E_k = \frac{1}{2}mv^2$$

Derived from: the work done to accelerate an object from rest to speed v equals $\frac{1}{2}mv^2$.

Gravitational Potential Energy

$$E_p = mgh$$

Where h is the vertical height gained. This applies near Earth's surface where g is approximately constant.

Conservation of Energy

Principle of Conservation of Energy: Energy cannot be created or destroyed, only transferred from one form to another. The total energy of a closed system remains constant.

For a falling object (no air resistance):

$$mgh = \frac{1}{2}mv^2 \quad \Rightarrow \quad v = \sqrt{(2gh)}$$

Power

Power: The rate of doing work or the rate of energy transfer.

$$P = W / t = E / t$$

Units: watts (W), where 1 W = 1 J s^{-1}.

For a constant force producing constant velocity:

$$P = Fv$$

This is useful for calculating the driving force or power of a vehicle at a given speed.

Efficiency

$$\text{Efficiency} = (\text{useful output energy} / \text{total input energy}) \times 100\%$$

$$\text{Efficiency} = (\text{useful output power} / \text{total input power}) \times 100\%$$

- Efficiency is always less than 100% in real systems (energy is dissipated as heat due to friction, air resistance, etc.)
- Efficiency has no units — it is a ratio (often expressed as a percentage)

Exam Tip: When a vehicle travels at constant velocity, the driving force equals the total resistive force (by Newton's First Law). The power output at this point is P = Fv = (resistive force) × v.

Worked Example: A car of mass 1200 kg travels at a constant 30 m s^{-1} on a level road. The resistive forces total 600 N. Calculate the power output of the engine.

At constant velocity, driving force = resistive force = 600 N (Newton's 1st Law)
P = Fv = 600 × 30 = **18 000 W = 18 kW**

Worked Example: A 0.15 kg ball is thrown at 20 m s^{-1} at a wall and rebounds at 15 m s^{-1}. The contact time is 0.030 s. Calculate the average force on the ball.

Taking the initial direction as positive: u = +20 m s^{-1}, v = −15 m s^{-1}
Δp = mv − mu = 0.15(−15) − 0.15(20) = −2.25 − 3.00 = −5.25 kg m s^{-1}
F = Δp/Δt = −5.25 / 0.030 = **−175 N**

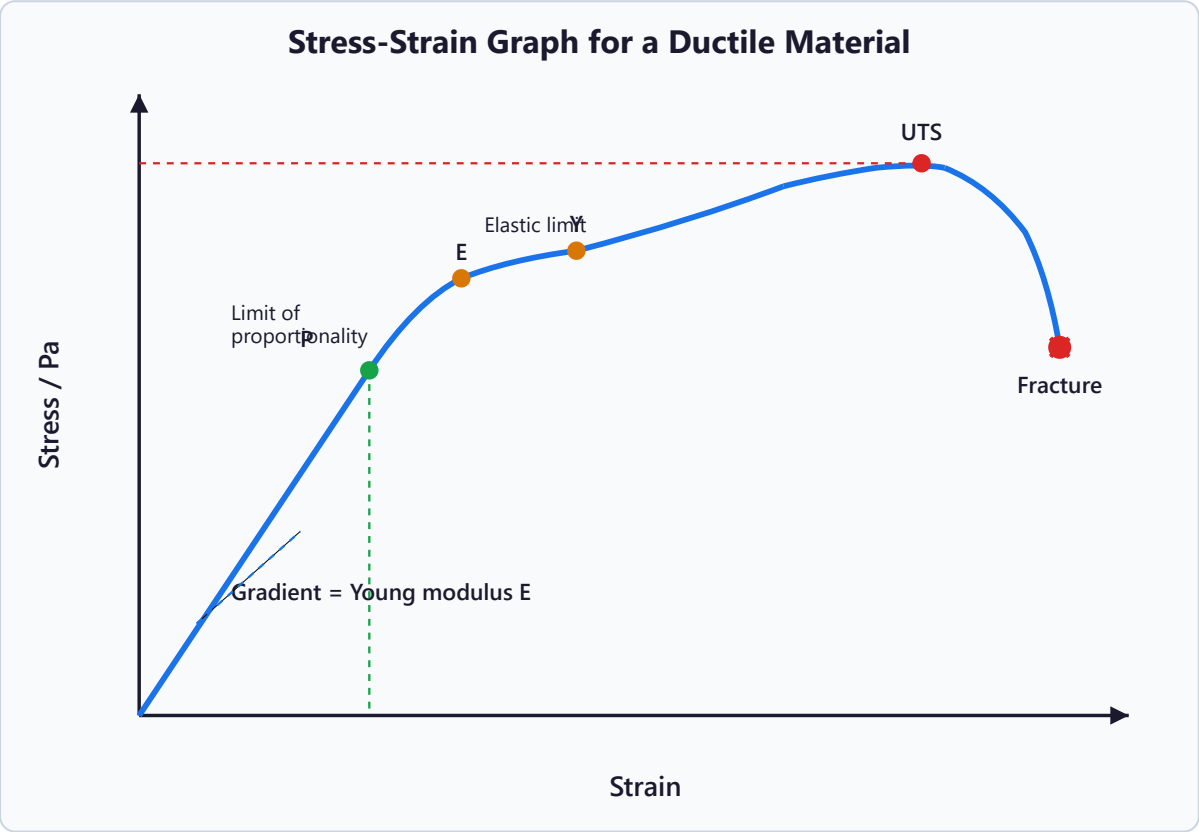
The magnitude is 175 N, directed away from the wall (negative = opposite to initial direction).

3.8 Materials — Stress, Strain, and Young Modulus

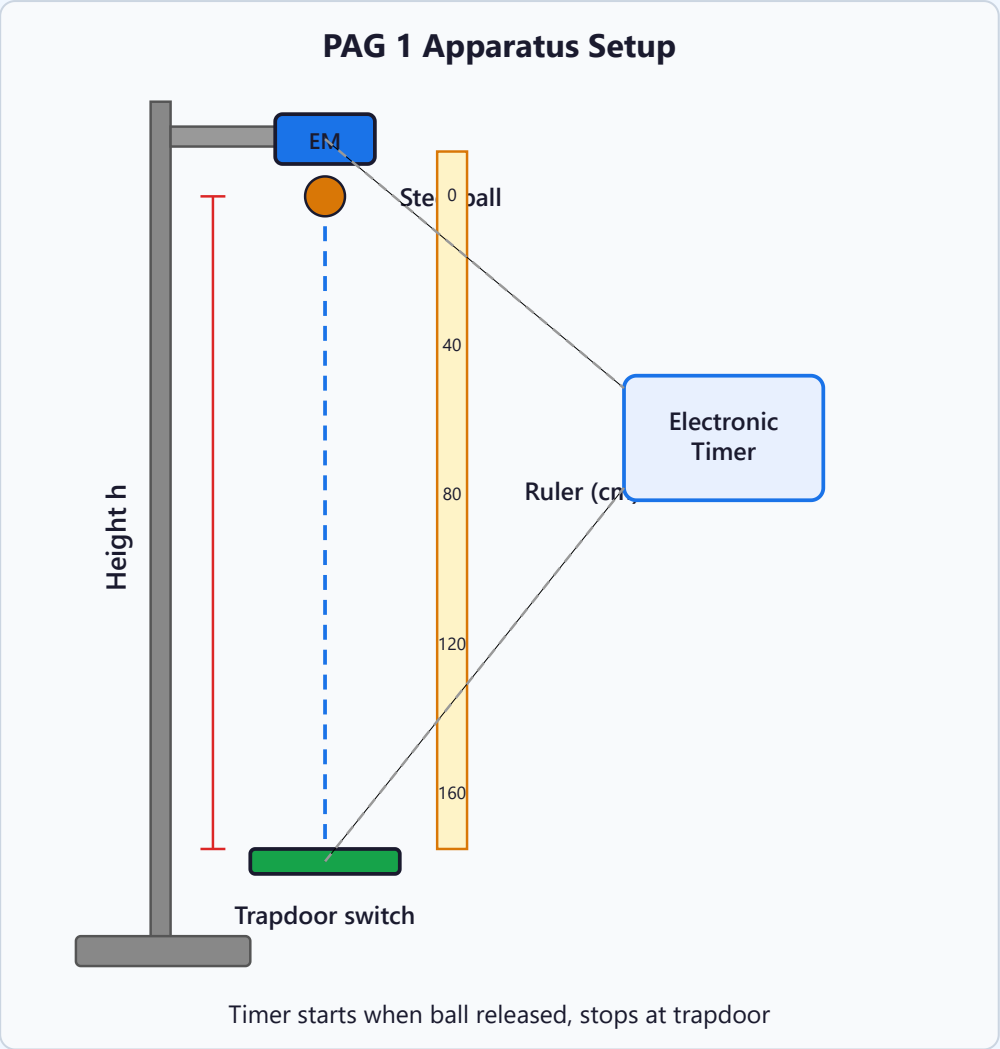
Stress (σ): Force per unit cross-sectional area. σ = F/A (Pa).

Strain (ε): Extension per unit original length. ε = ΔL/L (no units).

Young modulus (E): The ratio of stress to strain in the linear (elastic) region. E = σ/ε (Pa).



⚗ PAG 1: Determination of g by Free-Fall



Apparatus

- Electromagnet connected to a switch and low-voltage supply
- Steel ball bearing
- Trapdoor switch connected to an electronic timer (or datalogger)
- Metre ruler or 2-metre ruler (resolution 1 mm)
- Clamp stand, boss, and clamp

Method

1. Set up the electromagnet at the top of the clamp stand with the trapdoor directly below
2. Measure the height h from the bottom of the ball to the trapdoor using the ruler
3. Switch off the electromagnet to release the ball — the timer starts automatically and stops when the ball hits the trapdoor

4. Record the fall time t for this height
5. Repeat 3 times at each height and calculate the mean time
6. Vary h from about 0.2 m to 1.0 m in at least 6 intervals
7. Plot a graph of 2h (y-axis) against t^2 (x-axis)
8. The gradient of the straight line equals g

Safety

- Place a cushion or sand tray below the trapdoor to catch the ball
- Ensure the clamp stand has a heavy base to prevent toppling
- Keep feet clear of falling ball

Analysis

From $s = ut + ½at^2$, with u = 0: $h = ½gt^2$, so $2h = gt^2$. Plotting 2h vs t^2 gives a straight line through the origin with gradient = g.

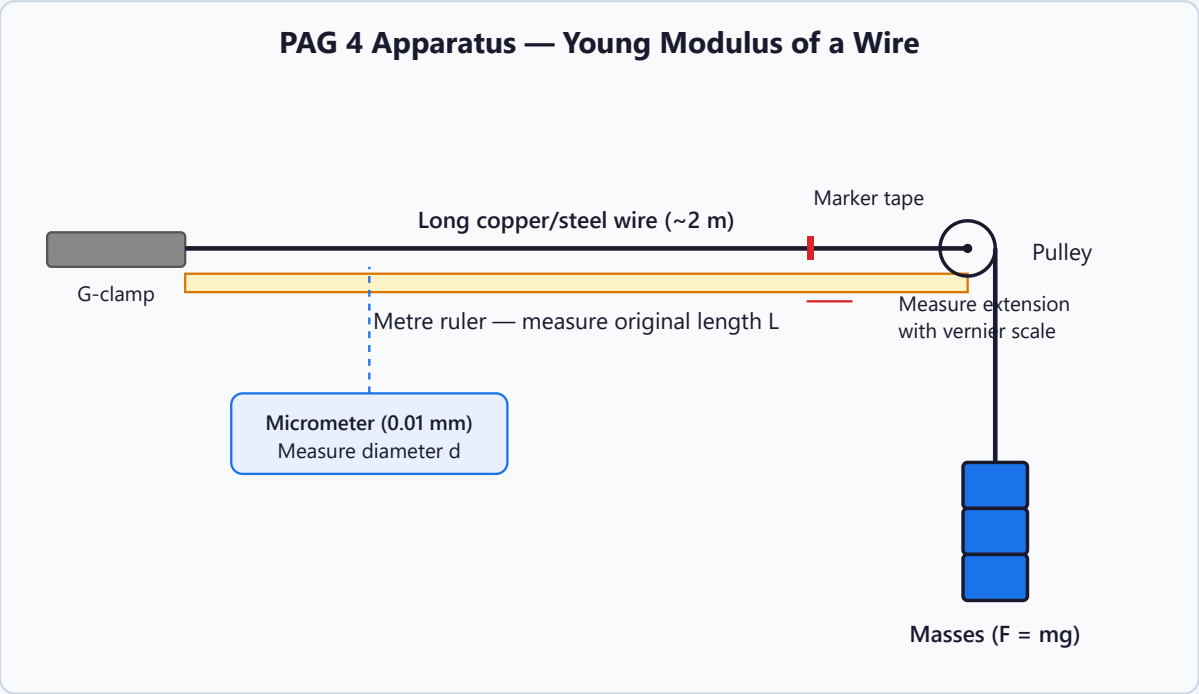
Sources of Error & Improvements

Source of Error	Type	Improvement
Reaction time in manual timing	Random	Use electromagnet + electronic timer (eliminates manual start)
Ball may not release instantly when EM is switched off (residual magnetism)	Systematic	Use a non-magnetic ball release mechanism or demagnetising circuit
Measuring h from ball centre to trapdoor	Systematic	Measure from bottom of ball; use set square for alignment
Air resistance on the ball	Systematic (makes g appear lower)	Use a small, dense ball to minimise drag

Exam Tip: The reason for plotting 2h vs t^2 (rather than h vs t) is to linearise the equation. The

gradient directly gives g without needing to halve it.

⚗ PAG 4: Determination of Young Modulus



Apparatus

- Long thin wire (~2 m) of copper or steel, clamped at one end
- Pulley clamped to the bench edge
- Mass hanger and slotted masses (e.g., 100 g increments)
- Metre ruler to measure original length L
- Micrometer screw gauge to measure the wire diameter d (at several points)
- Marker tape on wire with a vernier or ruler to measure extension ΔL
- Safety goggles

Method

1. Measure the original length L of the wire from the clamp to the marker using a metre ruler

2. Measure the diameter d of the wire at several points along its length using a micrometer, and calculate the mean. Calculate cross-sectional area $A = \pi d^2/4$
3. Add masses in increments (e.g., 100 g at a time, up to ~1 kg)
4. For each mass, record the extension ΔL from the marker position
5. Calculate stress ($\sigma = F/A$) and strain ($\varepsilon = \Delta L/L$) for each load
6. Plot a graph of stress (y-axis) against strain (x-axis)
7. The gradient of the linear region = Young modulus E
8. Check for elastic behaviour by removing masses and checking the wire returns to its original length

Safety

- Wear safety goggles — the wire may snap under tension
- Place a cushion or box below the masses to catch them if the wire breaks
- Do not exceed the elastic limit of the wire
- Keep feet clear of falling masses

Analysis

Young modulus E = stress / strain = gradient of the linear part of the stress-strain graph. Alternatively, $E = FL / A\Delta L$.

Sources of Error & Improvements

Source of Error	Type	Improvement
Wire diameter varies along its length	Random	Measure diameter at 6+ points along the wire and at different orientations; use the mean
Extension is very small (difficult to measure)	Random	Use a long wire (2+ m) to increase extension; use a vernier scale or travelling microscope
Wire may not be initially taut (kinks)	Systematic	Add a small initial load to straighten the wire before starting measurements

Thermal expansion of wire	Systematic	Perform experiment at constant room temperature; use a reference wire

Exam Tip: The biggest source of uncertainty is usually the diameter measurement (since $A = \pi d^2/4$, the percentage uncertainty in A is double that in d). Always take multiple readings of diameter.